COLLEGE EVENT MANAGEMENT SYSTEM

MongoDB NoSQL | Requirements Document

1. CRUD OPERATIONS

User Management

Create: Register new user account with email, password, name, phone, interests | Read: View user profile, event history, preferences | Update: Edit profile information, interests, and preferences | Delete: Deactivate user account

Event Management

Create: Publish new event with title, description, category, date, time, venue, capacity, poster image | Read: View event details, attendee list, feedback and ratings | Update: Edit event information, reschedule event, change status | Delete: Archive or remove event from system

Registration Management

Create: Student registers for event with RSVP confirmation | Read: View registration details, status, QR code for check-in | Update: Modify registration status to registered/waitlisted/checked-in | Delete: Unregister student from event

Venue Management

Create: Add new venue with location, capacity, amenities, images | Read: View venue details, availability, current bookings | Update: Modify venue information, capacity, amenities list | Delete: Archive or remove venue

Feedback Management

Create: Submit event feedback after attendance with rating and comments | Read: View all feedback, ratings, comments, sentiment analysis | Update: Edit submitted feedback and ratings | Delete: Remove feedback from system

Notification Management

Create: Generate notifications for event updates and reminders | Read: View notification history and content | Update: Mark notifications as read, archive notifications | Delete: Clear old or archived notifications

2. DATA VALIDATIONS

- Email: RFC 5322 format, unique across system, verified with confirmation link
- Phone Number: 10-15 digits, international format validation
- Event Title: 2-200 characters, must be unique per organizer per date
- Event Description: Minimum 50 characters, supports markdown formatting
- Event Dates and Times: Start date before end date, minimum 1 hour duration, future dates only
- Venue Capacity: Positive integer, registrations cannot exceed venue capacity

- Registration Status: Only valid values - registered, waitlisted, checked-in, cancelled, no-show
- Feedback Rating: Integer from 1 to 5 stars, optional text feedback up to 1000 characters
- Event Category: Must be one of - Academic, Cultural, Sports, Social, Workshop, Seminar
- Image Upload: JPG or PNG format only, maximum 5MB file size, auto-compressed to standard dimensions
- Organizer Verification: Organization must have valid registration number and current verification status
- Event Budget: Non-negative decimal value with detailed breakdown for events above threshold

3. ROLES & PERMISSIONS

Admin Role

- Full system access and control
- Manage all events - approve, reject, or remove events
- View all event analytics and detailed statistics
- Manage user accounts and organizer verification
- Enforce policies and compliance requirements
- Manage venue inventory and availability
- Export data and generate system-wide reports
- Monitor and manage system performance

User Role (Organizer)

- Create and publish events
- Manage event registrations and attendee list
- Track attendee check-ins and attendance
- View event-specific analytics and statistics
- Update event details and reschedule if needed
- Respond to student feedback and comments
- Generate reports for their events
- Promote events through platform

User Role (Student)

- Browse all available events on platform
- Search and filter events by category, date, location
- Register for events with RSVP confirmation
- View registered events and upcoming schedule
- Receive event reminders and notifications
- Provide ratings and feedback on attended events
- View event details and organizer information
- Connect with other attendees

4. REPORTS

Admin Reports:

- Event Statistics: Total events, active events, cancelled events, trending events
- Attendance Analytics: Attendance rate, no-show percentage, peak attendance hours
- Student Engagement: Event participation frequency, student interests, engagement metrics
- Organizer Performance: Events created, success rate, feedback ratings, compliance status
- Venue Utilization: Venue usage patterns, capacity optimization, booking trends
- System Activity: User registrations, login trends, system usage metrics

Organizer Reports:

- Event Analytics: Registration count, attendance rate, no-show count, feedback summary
- Attendee Report: Registered students, checked-in attendees, cancellations
- Feedback Analysis: Average ratings, common themes, sentiment analysis, suggestions
- Promotion Effectiveness: Views, clicks, registrations by traffic source

Student Reports:

- My Events: Registered events, attended events, upcoming events list
- Event History: Past events attended, ratings given, feedback submitted
- Personalized Recommendations: Suggested events based on interests and attendance history

5. FEEDBACK & SUGGESTIONS

Student Event Feedback:

Students submit post-event feedback using 1-5 star rating system. Feedback includes questions on event quality, organization, venue suitability, content relevance, and overall value. Open text field for detailed comments and improvement suggestions. System tracks feedback trends over time to identify patterns and common themes.

Organizer Feedback:

Admin provides feedback on event execution with 1-5 star rating. Includes compliance assessment, adherence to guidelines, strengths and areas for improvement. Comments on attendee satisfaction and recommendations for future events.

Platform Feedback & Suggestions:

All users can submit feature suggestions for platform improvements. Bug reporting system with error details and screenshot capabilities. Users recommend new features and improvements. Admin feedback dashboard displays feedback trends and common themes. Implementation tracking shows status of suggested features and improvements.